VIVEKANANDHA SAINIK COACHING CENTER

ALL INDIA SAINIK SCHOOL ENTRANCE EXAMINATION (AISSEE) 2026
CLASS IX MOCK PAPER 06

Time Allowed: 3 Hours (180 Mins) | Maximum Marks: 400 | Total Questions: 150

GENERAL INSTRUCTIONS:

1. This paper consists of 150 Multiple Choice Questions (MCQs) divided into 5 Sections:
   - **Section A: Mathematics** (Q. 1 to 50) — 4 Marks each (200 Marks)
   - **Section B: Intelligence** (Q. 51 to 75) — 2 Marks each (50 Marks)
   - **Section C: English** (Q. 76 to 100) — 2 Marks each (50 Marks)
   - **Section D: General Science** (Q. 101 to 125) — 2 Marks each (50 Marks)
   - **Section E: Social Science** (Q. 126 to 150) — 2 Marks each (50 Marks)
2. All questions are compulsory. There is NO negative marking for wrong answers.
3. Mark your responses clearly on the OMR sheet. Select exactly ONE correct option for each question.

PART A: QUESTION PAPER (Q. 1 TO 150)

SECTION A: MATHEMATICS (Q. 1 - 50) [50 Qs x 4 Marks = 200 Marks]

Q.1. What should be added to -5/8 to 3/4 ?

(a) 11/8 **(b)** 1/8
(c) -11/8 **(d)** 7/8

Q.2. Solve for x: (2x + 1) / (3x - 2) = 5/9.

(a) -19/3 **(b)** 19/3
(c) -3/19 **(d)** -5

Q.3. Three angles of a quadrilateral are 75°, 90°, and 110°. Find the measure of the fourth angle.

(a) 75° **(b)** 80°
(c) 85° **(d)** 95°

Q.4. Find the value of √17.64.

(a) 4.1 **(b)** 4.2
(c) 4.4 **(d)** 4.8

Q.5. Find the volume of a cube whose total surface area is 294 cm².

(a) 216 cm³ **(b)** 343 cm³
(c) 512 cm³ **(d)** 729 cm³

Q.6. Evaluate: (2^(-1) × 4^(-1)) ÷ 2^(-2).

(a) 1/4 **(b)** 1/2
(c) 1 **(d)** 2

Q.7. A laptop was bought for 40,000. Its value depreciates at 10% per annum. What will be its value after 2 years?

(a) 32,000 **(b)** 32,400
(c) 36,000 **(d)** 34,200

Q.8. Find the compound interest on 5000 for 2 years at 8% per annum, compounded annually.

(a) 800 **(b)** 832
(c) 864 **(d)** 900

Q.9. If x - y = 7 and xy = 9, find the value of x^2 + y^2.

(a) 49 **(b)** 58
(c) 67 **(d)** 85

Q.10. Factorise: $6x^2 + 11x + 3$.

(a) (3x + 1)(2x + 3) **(b)** (6x + 1)(x + 3)
(c) (2x + 1)(3x + 3) **(d)** (3x - 1)(2x - 3)

Q.11. A car travels 180 km in 3 hours at uniform speed. How far will it travel in 5 hours at the same speed?

(a) 250 km **(b)** 280 km
(c) 300 km **(d)** 320 km

Q.12. The area of a trapezium is 240 cm^2 and its height is 12 cm. If one parallel side is 16 cm, find the other parallel side.

(a) 20 cm **(b)** 22 cm
(c) 24 cm **(d)** 28 cm

Q.13. A cylindrical tank has a radius of 14 cm and a height of 5 cm. Its volume is (Use $\pi = 22/7$):

(a) 1540 cm^3 **(b)** 3080 cm^3
(c) 2200 cm^3 **(d)** 4400 cm^3

Q.14. A standard die is thrown once. What is the probability of getting an even prime number?

(a) 1/2 **(b)** 1/3
(c) 1/6 **(d)** 2/3

Q.15. If a polyhedron has 6 vertices and 12 edges, find the number of faces it has.

(a) 6 **(b)** 7
(c) 8 **(d)** 10

Q.16. The ratio of length to breadth of a rectangular field is 5 : 3. If its perimeter is 320 m, find its area.

(a) 4800 m^2 **(b)** 5400 m^2
(c) 6000 m^2 **(d)** 6400 m^2

Q.17. The diagonals of a rhombus are 24 cm and 10 cm. Its area and side length are respectively:

(a) 120 cm^2, 13 cm **(b)** 240 cm^2, 13 cm
(c) 120 cm^2, 17 cm **(d)** 180 cm^2, 15 cm

Q.18. Find the smallest square number divisible by each of the numbers 6, 9, and 15.

(a) 180 **(b)** 360
(c) 900 **(d)** 1600

Q.19. Divide (a^2 - b^2) by (a + b).

(a) a + b **(b)** a - b
(c) a^2 + b^2 **(d)** ab

Q.20. An article marked at ■1500 is sold at a 20% discount. Find the selling price.

(a) ■1100 **(b)** ■1200
(c) ■1250 **(d)** ■1300

Q.21. If a 4-digit number 21y5 is divisible by 9, find the digit y.

(a) 0 **(b)** 1
(c) 2 **(d)** 3

Q.22. The distance between Earth and Sun is approximately 149,600,000,000 m. In standard scientific notation, this is:

(a) 1.496 × 10^10 m **(b)** 1.496 × 10^11 m
(c) 14.96 × 10^10 m **(d)** 1.496 × 10^12 m

Q.23. The total surface area of a cuboid of length 10 cm, breadth 8 cm, and height 6 cm is:

(a) 188 cm² **(b)** 376 cm²
(c) 480 cm² **(d)** 240 cm²

Q.24. Two coins are tossed simultaneously. What is the probability of getting AT LEAST one head?

(a) 1/4 **(b)** 1/2
(c) 3/4 **(d)** 1

Q.25. Three consecutive integers add up to 51. Find the smallest integer.

(a) 15 **(b)** 16
(c) 17 **(d)** 18

Q.26. How many zeros will be at the end of the square of 400?

(a) 2 **(b)** 3
(c) 4 **(d)** 6

Q.27. 6 men can complete a job in 12 days. In how many days can 9 men complete the same job?

(a) 6 days **(b)** 8 days
(c) 9 days **(d)** 10 days

Q.28. Add $(7x^2 - 4x + 5)$ and $(9x - 10)$.

(a) $7x^2 + 5x - 5$ **(b)** $7x^2 - 13x + 15$
(c) $16x^2 + 5x - 5$ **(d)** $7x^2 + 5x + 5$

Q.29. What is the additive inverse of -11/19?

(a) 19/11 **(b)** 11/19
(c) -19/11 **(d)** 1

Q.30. What is the sum of all interior angles of a regular hexagon?

(a) 540° **(b)** 720°
(c) 900° **(d)** 1080°

Q.31. An article bought for ■500 is sold for ■575. The gain percentage is:

(a) 12% **(b)** 15%
(c) 18% **(d)** 20%

Q.32. What is the smallest number by which 256 must be divided to get a perfect cube?

(a) 2 **(b)** 4
(c) 8 **(d)** 16

Q.33. What is the abscissa of the point (3, -5)?

(a) 3 **(b)** -5
(c) -3 **(d)** 5

Q.34. Find the area of a circular plate of diameter 28 cm (Use $\pi = 22/7$).

(a) 308 cm² **(b)** 616 cm²
(c) 1232 cm² **(d)** 88 cm²

Q.35. Factorise: $25x^2 - 49y^2$.

(a) $(5x - 7y)(5x + 7y)$ **(b)** $(25x - 49y)(x + y)$
(c) $(5x - 7y)^2$ **(d)** $(5x + 7y)^2$

Q.36. Value of (2/3)^(-2) is:

(a) 4/9 **(b)** -4/9
(c) 9/4 **(d)** -9/4

Q.37. Which of the following numbers is divisible by 11?

(a) 1234 **(b)** 1331
(c) 1442 **(d)** 1554

Q.38. Marked Price = ■800, Discount = 10%. The Selling Price is:

(a) ■700 **(b)** ■720
(c) ■740 **(d)** ■760

Q.39. Solve: x/2 - 1/5 = x/3 + 1/4.

(a) 2.1 **(b)** 2.5
(c) 2.7 **(d)** 3.2

Q.40. If the area of a rhombus is 120 cm^2 and one diagonal is 10 cm, find the length of the other diagonal.

(a) 18 cm **(b)** 20 cm
(c) 24 cm **(d)** 28 cm

Q.41. Area of trapezium = 180 cm^2, parallel sides are 18 cm and 12 cm. Distance between them is:

(a) 10 cm **(b)** 12 cm
(c) 14 cm **(d)** 15 cm

Q.42. Find the amount on ■2000 at 10% per annum for 2 years, compounded annually.

(a) ■2200 **(b)** ■2400
(c) ■2420 **(d)** ■2440

Q.43. Simplify: (5^2)^3 ÷ 5^4.

(a) 5 **(b)** 10
(c) 25 **(d)** 125

Q.44. Which of the following CANNOT be the units digit of a perfect square?

(a) 1 **(b)** 2
(c) 5 **(d)** 9

Q.45. What is the degree of the polynomial $3x^3 - 5x^2 + 7x - 2$?

(a) 1 **(b)** 2
(c) 3 **(d)** 4

Q.46. Find the range of the data set: 12, 25, 18, 30, 8, 22.

(a) 18 **(b)** 20
(c) 22 **(d)** 25

Q.47. The volume of a cylinder of height 10 cm is 1540 cm^3. Find its radius (Use $\pi = 22/7$).

(a) 5 cm **(b)** 6 cm
(c) 7 cm **(d)** 14 cm

Q.48. Speed and time taken to cover a fixed distance are in:

(a) Direct proportion **(b)** Inverse proportion
(c) Neither **(d)** Both

Q.49. Value of (x + 4)(x + 10) using identity (x + a)(x + b) = x² + (a + b)x + ab is:

(a) x² + 14x + 40 **(b)** x² + 40x + 14
(c) x² + 14x + 14 **(d)** x² + 40

Q.50. Is multiplication distributive over addition for rational numbers?

(a) Yes, always **(b)** No, never
(c) Only for positive numbers **(d)** Only for integers

SECTION B: INTELLIGENCE (Q. 51 - 75) [25 Qs x 2 Marks = 50 Marks]

Q.51. Doctor : Patient :: Teacher : ?

(a) School **(b)** Student
(c) Book **(d)** Class

Q.52. Find the missing term: 2, 6, 12, 20, 30, ?

(a) 40 **(b)** 42
(c) 44 **(d)** 46

Q.53. If 'BOMBAY' is coded as 'CPNCBZ', how is 'DELHI' coded?

(a) EFMIJ **(b)** CDKGH
(c) FGMJK **(d)** EDMHJ

Q.54. Choose the odd one out:

(a) Copper **(b)** Zinc
(c) Iron **(d)** Wood

Q.55. A is B's sister. C is B's mother. D is C's father. How is A related to D?

(a) Daughter **(b)** Granddaughter
(c) Sister **(d)** Niece

Q.56. A man walks 10 km East, turns South and walks 5 km, then turns West and walks 10 km. How far is he from starting point?

(a) 5 km **(b)** 10 km
(c) 15 km **(d)** 20 km

Q.57. What comes next in letter series: Z, X, V, T, R, ?

(a) O **(b)** P
(c) Q **(d)** N

Q.58. In a queue of 30 people, Priya is 12th from the front. What is her position from the end?

(a) 18th **(b)** 19th
(c) 20th **(d)** 21st

Q.59. Complete analogy: ACE : GIK :: MOQ : ?

(a) SUW **(b)** RTV
(c) TVX **(d)** PRT

Q.60. If 'P' means '+', 'Q' means '-', 'R' means '×', and 'S' means '÷', find the value of: 18 R 12 S 4 P 5 Q 6.

(a) 51 **(b)** 53
(c) 55 **(d)** 57

Q.61. A die is rolled. What is the probability of getting a number greater than 4?

(a) 1/6 **(b)** 1/3
(c) 1/2 **(d)** 2/3

Q.62. Choose the correct mirror image of '581' facing vertical mirror on right.

(a) 185 **(b)** 181
(c) 581 **(d)** 185

Q.63. Find the missing number: Row 1: [3, 4, 25], Row 2: [5, 12, 169], Row 3: [8, 15, ?].

(a) 225 **(b)** 289
(c) 300 **(d)** 324

Q.64. Six people P, Q, R, S, T, U sit in a circle facing center. P is opposite R. Q is to immediate right of P. T sits to the immediate left of R, Who sits opposite to Q?

(a) S **(b)** T
(c) U **(d)** R

Q.65. Which Venn diagram represents: Stationary, Pencils, Erasers?

(a) Two disjoint circles inside a large circle **(b)** Three intersecting circles
(c) Three concentric circles **(d)** Separate circles

Q.66. Choose the odd one out:

(a) January **(b)** March
(c) April **(d)** May

Q.67. Complete analogy: 8 : 64 :: 10 : ?

(a) 80 **(b)** 90
(c) 100 **(d)** 110

Q.68. If 'CAT' = 24 and 'DOG' = 26, then 'PIG' = ?

(a) 30 **(b)** 32
(c) 34 **(d)** 36

Q.69. The angle between clock hands at 9:00 AM is:

(a) 45° **(b)** 90°
(c) 120° **(d)** 180°

Q.70. How many squares are there in a 2 × 2 grid?

(a) 4 **(b)** 5
(c) 6 **(d)** 8

Q.71. Which word CAN be formed using letters of 'PREPARATION'?

(a) NATIONAL **(b)** PAPER
(c) PARTITION **(d)** ACTION

Q.72. Complete the number series: 1, 4, 9, 16, 25, 36, ?

(a) 42 **(b)** 45
(c) 49 **(d)** 54

Q.73. Logical order: 1. Member, 2. Family, 3. Community, 4. Nation, 5. Society.

(a) 1, 2, 3, 5, 4 **(b)** 1, 2, 4, 3, 5
(c) 2, 1, 3, 5, 4 **(d)** 1, 3, 2, 5, 4

Q.74. Statements: Some books are pens. All pens are pencils. Conclusions: I. Some books are pencils. II. All pencils are books.

(a) Only conclusion I follows **(b)** Only II follows
(c) Both follow **(d)** Neither follows

Q.75. What is the water image of the letter 'E'?

(a) E **(b)** 3
(c) e **(d)** F

SECTION C: ENGLISH (Q. 76 - 100) [25 Qs x 2 Marks = 50 Marks]

Q.76. Who is regarded as the chief architect of the Indian Constitution?

(a) Mahatma Gandhi **(b)** Dr. B.R. Ambedkar
(c) Jawaharlal Nehru **(d)** Dr. Rajendra Prasad

Q.77. Dr. Ambedkar served as the Chairman of which committee?

(a) Advisory Committee **(b)** Steering Committee
(c) Drafting Committee **(d)** Finance Committee

Q.78. What is defined as the supreme law of India in the passage?

(a) Penal Code **(b)** Constitution of India
(c) Parliamentary Gazette **(d)** Supreme Court Directives

Q.79. Which values are explicitly mentioned in the passage as assured to citizens?

(a) Absolute monarchy **(b)** Justice, equality, and liberty
(c) Feudal hierarchy **(d)** Military dominance

Q.80. In the passage, which word means 'essential' or 'primary'?

(a) Fundamental **(b)** Architecture
(c) Committee **(d)** Procedure

Q.81. Fill in the blank: She _____ English for two hours before the guests arrived.

(a) had been studying **(b)** is studying
(c) studies **(d)** will study

Q.82. Change to Passive Voice: 'Who wrote this famous book?'

(a) By whom was this famous book written? **(b)** Who was written this book?
(c) By whom this book wrote? **(d)** Whom wrote this book?

Q.83. Change to Indirect Speech: The teacher said, 'The Earth revolves around the Sun.'

(a) The teacher said that the Earth revolved around the Sun. **(b)** The teacher said that the Earth revolves around the Sun.
(c) The teacher told that Earth was revolving. **(d)** The teacher says Earth revolves.

Q.84. Fill in the blanks: He is _____ honest man who lives in _____ small village.

(a) an, a **(b)** a, an
(c) an, an **(d)** a, a

Q.85. Fill in the blank: Distribute these sweets _____ the four children.

(a) between **(b)** among
(c) in **(d)** with

Q.86. Fill in the blank: She worked hard, _____ she failed to secure first rank.

(a) yet **(b)** so
(c) because **(d)** and

Q.87. Choose the correct modal: _____ I come in, Sir?

(a) May **(b)** Might
(c) Would **(d)** Should

Q.88. Fill in the blank: Bread and butter _____ his favourite breakfast.

(a) is **(b)** are
(c) were **(d)** have been

Q.89. Choose the question tag: You haven't finished your homework yet, _____?

(a) have you **(b)** haven't you
(c) do you **(d)** did you

Q.90. Choose the word SIMILAR in meaning to 'COURTEOUS':

(a) Rude **(b)** Polite
(c) Harsh **(d)** Proud

Q.91. Choose the word OPPOSITE in meaning to 'ANCIENT':

(a) Old **(b)** Modern
(c) Historic **(d)** Primitive

Q.92. One-word substitution: A remedy for all diseases or difficulties:

(a) Panacea **(b)** Antidote
(c) Placebo **(d)** Vaccine

Q.93. What does the idiom 'Once in a blue moon' mean?

(a) Frequently **(b)** Very rarely
(c) Every month **(d)** In the evening

Q.94. Choose the correctly spelt word:

(a) Recieve **(b)** Receive
(c) Receve **(d)** Recievee

Q.95. Spot the error: 'The news (A) / are (B) / very encouraging (C) / today (D).'

(a) Part A **(b)** Part B
(c) Part C **(d)** Part D

Q.96. Rearrange P, Q, R, S: P: health is, Q: wealth, R: Prevention is, S: better than cure.

(a) RSQP **(b)** PQRS
(c) RSPQ **(d)** QPSR

Q.97. A group of ships sailing together is called a:

(a) Fleet **(b)** Herd
(c) Flight **(d)** Swarm

Q.98. Change to Positive Degree: 'Iron is more useful than any other metal.'

(a) No other metal is as useful as iron. **(b)** Iron is the most useful metal.
(c) Very few metals are as useful as iron. **(d)** Iron is useful metal.

Q.99. What is the masculine form of 'Spinster'?

(a) Bachelor **(b)** Wizard
(c) Drake **(d)** Monk

Q.100. What does the phrase 'By leaps and bounds' mean?

(a) Very slowly **(b)** Rapidly
(c) Secretly **(d)** Irregularly

SECTION D: GENERAL SCIENCE (Q. 101 - 125) [25 Qs x 2 Marks = 50 Marks]

Q.101. Which force always opposes relative motion between two surfaces in contact?

(a) Gravitational force **(b)** Frictional force
(c) Electrostatic force **(d)** Magnetic force

Q.102. Atmospheric pressure is measured using a:

(a) Thermometer **(b)** Barometer
(c) Hydrometer **(d)** Lactometer

Q.103. Lubricants are substances applied between machine parts to:

(a) Increase friction **(b)** Reduce friction
(c) Increase weight **(d)** Prevent motion

Q.104. Sound wave CANNOT travel through:

(a) Solids **(b)** Liquids
(c) Gases **(d)** Vacuum

Q.105. Loudness of sound is directly proportional to the square of its:

(a) Frequency **(b)** Amplitude
(c) Speed **(d)** Time period

Q.106. Which of the following liquids is a GOOD conductor of electricity?

(a) Distilled water **(b)** Lemon juice
(c) Vegetable oil **(d)** Sugar solution

Q.107. The phenomenon of splitting of white light into seven constituent colors is called:

(a) Reflection **(b)** Refraction
(c) Dispersion **(d)** Diffraction

Q.108. The eye defect where a person cannot see nearby objects clearly is called:

(a) Myopia **(b)** Hypermetropia
(c) Cataract **(d)** Glaucoma

Q.109. The largest planet in our solar system is:

(a) Saturn **(b)** Jupiter
(c) Neptune **(d)** Uranus

Q.110. An instrument used to detect whether an object carries an electric charge is:

(a) Periscope **(b)** Electroscope
(c) Telescope **(d)** Kaleidoscope

Q.111. Which synthetic polymer is used to make non-stick coating on cookware?

(a) PVC **(b)** Teflon (PTFE)
(c) Bakelite **(d)** Polystyrene

Q.112. Which metal reacts with dilute hydrochloric acid releasing hydrogen gas with a 'pop' sound?

(a) Copper **(b)** Zinc
(c) Silver **(d)** Gold

Q.113. Which non-metal is a good conductor of electricity?

(a) Diamond **(b)** Graphite
(c) Sulphur **(d)** Iodine

Q.114. The cleanest and least polluting gaseous fossil fuel used in automobiles is:

(a) LPG **(b)** CNG
(c) Coal gas **(d)** Diesel

Q.115. The innermost zone of a candle flame is the:

(a) Hottest zone **(b)** Dark zone (least hot)
(c) Luminous zone **(d)** Blue zone

Q.116. Depletion of the protective ozone layer in the atmosphere is caused by:

(a) CO_2 **(b)** Chlorofluorocarbons (CFCs)
(c) SO_2 **(d)** Methane

Q.117. The process of separating grain seeds from the chaff after harvesting is called:

(a) Sowing **(b)** Threshing
(c) Tilling **(d)** Weeding

Q.118. The world's first antibiotic Penicillin was discovered by:

(a) Louis Pasteur **(b)** Alexander Fleming
(c) Edward Jenner **(d)** Robert Koch

Q.119. Species found exclusively in a specific geographic area are termed:

(a) Extinct species **(b)** Endemic species
(c) Endangered species **(d)** Exotic species

Q.120. The green pigment in plant leaves essential for trapping solar energy is:

(a) Haemoglobin **(b)** Chlorophyll
(c) Xanthophyll **(d)** Carotene

Q.121. External fertilization occurs commonly in:

(a) Hens and Humans **(b)** Frogs and Fish
(c) Dogs and Cats **(d)** Cows and Goats

Q.122. The master endocrine gland that controls other endocrine glands is:

(a) Thyroid **(b)** Pituitary gland
(c) Adrenal **(d)** Pancreas

Q.123. Which organelle is famously known as the 'Powerhouse of the Cell'?

(a) Nucleus **(b)** Mitochondria
(c) Ribosome **(d)** Lysosome

Q.124. Rusting of iron requires the essential presence of both:

(a) Water and Nitrogen **(b)** Oxygen and Water (moisture)
(c) CO_2 and Light **(d)** Oxygen and Oil

Q.125. The vaccine for Smallpox was discovered by:

(a) Edward Jenner **(b)** Louis Pasteur
(c) Robert Hooke **(d)** Anton van Leeuwenhoek

SECTION E: SOCIAL SCIENCE (Q. 126 - 150) [25 Qs x 2 Marks = 50 Marks]

Q.126. The decisive Battle of Buxar was fought in the year:

(a) 1757 **(b)** 1764
(c) 1776 **(d)** 1857

Q.127. Who was proclaimed Emperor of India by the rebel sepoys in Delhi during 1857 Uprising?

(a) Bahadur Shah Zafar **(b)** Nana Saheb
(c) Rani Laxmibai **(d)** Bakht Khan

Q.128. Who initiated the Young Bengal Movement in the 1820s?

(a) Henry Louis Vivian Derozio **(b)** Ishwar Chandra Vidyasagar
(c) Swami Dayanand **(d)** Keshab Chandra Sen

Q.129. The tragic Jallianwala Bagh massacre took place on 13th April 1919 at:

(a) Lahore **(b)** Amritsar
(c) Delhi **(d)** Jalandhar

Q.130. Who proclaimed the slogan 'Swaraj is my birthright and I shall have it'?

(a) Lala Lajpat Rai **(b)** Bal Gangadhar Tilak
(c) Bipin Chandra Pal **(d)** Gopal Krishna Gokhale

Q.131. Renewable resources are those natural resources which:

(a) Have strictly limited stock **(b)** Get renewed or replenished quickly by natural processes
(c) Are artificially manufactured **(d)** Take millions of years to form

Q.132. Contour ploughing on hill slopes is done to:

(a) Increase water runoff **(b)** Decelerate water flow down slopes and check soil erosion
(c) Trap forest animals **(d)** Grow teak trees

Q.133. Which variety of coal has the highest carbon content and quality?

(a) Lignite **(b)** Bituminous
(c) Anthracite **(d)** Peat

Q.134. Commercial rearing of silkworms for raw silk production is known as:

(a) Pisciculture **(b)** Viticulture
(c) Sericulture **(d)** Horticulture

Q.135. Which Indian city is popularly known as the 'Manchester of India'?

(a) Mumbai **(b)** Ahmedabad
(c) Kanpur **(d)** Coimbatore

Q.136. The Preamble to the Indian Constitution declares India to be a:

(a) Sovereign, Socialist, Secular, Democratic Republic **(b)** Federal Monarchy
(c) Capitalist Democracy **(d)** Unitary State

Q.137. Which Fundamental Right was called the 'Heart and Soul of the Constitution' by Dr. Ambedkar?

(a) Right to Equality **(b)** Right to Freedom
(c) Right to Constitutional Remedies (Art. 32) **(d)** Right to Education

Q.138. Members of the Rajya Sabha are elected for a term of:

(a) 4 years **(b)** 5 years
(c) 6 years **(d)** 2 years

Q.139. What is the highest judicial court of appeal in India?

(a) High Court **(b)** District Court
(c) Supreme Court of India **(d)** Sessions Court

Q.140. The Minimum Wages Act enacted by the government ensures that:

(a) Workers receive wages not lower than a legally fixed minimum **(b)** Workers get free housing
(c) Pension is paid at age 50 **(d)** Medical care is free

Q.141. The Ryotwari System of land revenue was introduced in South India by:

(a) Lord Cornwallis **(b)** Thomas Munro and Alexander Read
(c) Holt Mackenzie **(d)** Warren Hastings

Q.142. Birsa Munda led a historic tribal rebellion against British rule in:

(a) Santhal Parganas **(b)** Chota Nagpur region
(c) Bastar **(d)** Khasi Hills

Q.143. Natural gas is frequently found in geological association with:

(a) Coal seams **(b)** Petroleum deposits
(c) Bauxite reserves **(d)** Iron ore mines

Q.144. The difference between the Crude Birth Rate and Crude Death Rate of a country is called its:

(a) Natural growth rate **(b)** Migration rate
(c) Density index **(d)** Dependency ratio

Q.145. Which authority holds primary responsibility for providing essential public facilities like water and sanitation?

(a) Private corporations **(b)** Government sector
(c) Foreign investors **(d)** Individual contractors

Q.146. Arya Samaj was founded in 1875 by:

(a) Swami Dayanand Saraswati **(b)** Ramakrishna Paramahamsa
(c) Ishwar Chandra Vidyasagar **(d)** Debendranath Tagore

Q.147. Which mineral is the chief ore used for extracting Aluminum metal?

(a) Hematite **(b)** Bauxite
(c) Magnetite **(d)** Copper pyrite

Q.148. Who is the Constitutional Head of the Executive of the Union of India?

(a) Prime Minister **(b)** President of India
(c) Chief Justice of India **(d)** Cabinet Secretary

Q.149. The historic 'Quit India' resolution was adopted by Indian National Congress in August 1942 at:

(a) Bombay (Gowalia Tank) **(b)** Calcutta
(c) Lahore **(d)** Madras

Q.150. Article 21A was added to the Constitution to grant the Fundamental Right to:

(a) Clean Environment **(b)** Free and Compulsory Education for children aged 6 to 14
(c) Employment **(d)** Property

PART B: MASTER ANSWER KEY (Q. 1 TO 150)

Q.	Ans	Q.	Ans	Q.	Ans	Q.	Ans	Q.	Ans
1	(a)	31	(b)	61	(b)	91	(b)	121	(b)
2	(a)	32	(b)	62	(d)	92	(a)	122	(b)
3	(c)	33	(a)	63	(b)	93	(b)	123	(b)
4	(b)	34	(b)	64	(b)	94	(b)	124	(b)
5	(b)	35	(a)	65	(a)	95	(b)	125	(a)
6	(b)	36	(c)	66	(c)	96	(c)	126	(b)
7	(b)	37	(b)	67	(c)	97	(a)	127	(a)
8	(b)	38	(b)	68	(b)	98	(a)	128	(a)
9	(c)	39	(c)	69	(b)	99	(a)	129	(b)
10	(a)	40	(c)	70	(b)	100	(b)	130	(b)
11	(c)	41	(b)	71	(b)	101	(b)	131	(b)
12	(c)	42	(c)	72	(c)	102	(b)	132	(b)
13	(b)	43	(c)	73	(a)	103	(b)	133	(c)
14	(c)	44	(b)	74	(a)	104	(d)	134	(c)
15	(c)	45	(c)	75	(a)	105	(b)	135	(b)
16	(c)	46	(c)	76	(b)	106	(b)	136	(a)
17	(a)	47	(c)	77	(c)	107	(c)	137	(c)
18	(c)	48	(b)	78	(b)	108	(b)	138	(c)
19	(b)	49	(a)	79	(b)	109	(b)	139	(c)
20	(b)	50	(a)	80	(a)	110	(b)	140	(a)
21	(b)	51	(b)	81	(a)	111	(b)	141	(b)
22	(b)	52	(b)	82	(a)	112	(b)	142	(b)
23	(b)	53	(a)	83	(b)	113	(b)	143	(b)
24	(c)	54	(d)	84	(a)	114	(b)	144	(a)
25	(b)	55	(b)	85	(b)	115	(b)	145	(b)
26	(c)	56	(a)	86	(a)	116	(b)	146	(a)
27	(b)	57	(b)	87	(a)	117	(b)	147	(b)
28	(a)	58	(b)	88	(a)	118	(b)	148	(b)
29	(b)	59	(a)	89	(a)	119	(b)	149	(a)
30	(b)	60	(b)	90	(b)	120	(b)	150	(b)

PART C: DETAILED STEP-BY-STEP SOLUTIONS (Q. 1 TO 150)

SECTION A: MATHEMATICS (Q. 1 - 50) [50 Qs x 4 Marks = 200 Marks]

Question 1: Correct Option (a)

-&frac5;8 + x = &frac3;4 ⇒ x = &frac3;4 + &frac5;8 = (6 + 5)/8 = 11/8.

Question 2: Correct Option (a)

Cross multiply: 9(2x + 1) = 5(3x - 2) ⇒ 18x + 9 = 15x - 10 ⇒ 3x = -19 ⇒ x = -19/3.

Question 3: Correct Option (c)

Sum of interior angles of a quadrilateral = 360°. 75° + 90° + 110° + x = 360° ⇒ 275° + x = 360° ⇒ x = 85°.

Question 4: Correct Option (b)

17.64 = 1764 / 100. √1764 = 42. So √17.64 = 42 / 10 = 4.2.

Question 5: Correct Option (b)

TSA of cube = 6a^2 = 294 ⇒ a^2 = 49 ⇒ side a = 7 cm. Volume = a^3 = 7^3 = 343 cm³.

Question 6: Correct Option (b)

(1/2 × 1/4) ÷ (1/4) = (1/8) ÷ (1/4) = (1/8) × 4 = 1/2.

Question 7: Correct Option (b)

Depreciated Value = 40000(1 - 10/100)^2 = 40000(0.9)^2 = 40000 × 0.81 = ■32,400.

Question 8: Correct Option (b)

Amount A = 5000(1 + 8/100)^2 = 5000(1.08)^2 = 5000 × 1.1664 = ■5832. CI = 5832 - 5000 = ■832.

Question 9: Correct Option (c)

Using (x - y)^2 = x^2 + y^2 - 2xy: 7^2 = x^2 + y^2 - 2(9) ⇒ 49 = x^2 + y^2 - 18 ⇒ x^2 + y^2 = 67.

Question 10: Correct Option (a)

Split the middle term: 6x² + 9x + 2x + 3 = 3x(2x + 3) + 1(2x + 3) = (3x + 1)(2x + 3).

Question 11: Correct Option (c)

Speed = Distance / Time = 180 / 3 = 60 km/h. Distance in 5 hours = 60 × 5 = 300 km.

Question 12: Correct Option (c)

Area = &frac1;2 × (a + b) × h ⇒ 240 = &frac1;2 × (16 + b) × 12 ⇒ 6(16 + b) = 240 ⇒ 16 + b = 40 ⇒ b = 24 cm.

Question 13: Correct Option (b)

Volume = πr²h = (22/7) × 14 × 14 × 5 = 22 × 28 × 5 = 3080 cm³.

Question 14: Correct Option (c)

Outcomes on a die = {1, 2, 3, 4, 5, 6}. The only even prime number is 2 (1 outcome). P = 1/6.

Question 15: Correct Option (c)

By Euler's formula F + V - E = 2 ⇒ F + 6 - 12 = 2 ⇒ F - 6 = 2 ⇒ F = 8 faces.

Question 16: Correct Option (c)

Perimeter = 2(5x + 3x) = 16x = 320 ⇒ x = 20. Length = 100 m, Breadth = 60 m. Area = 100 × 60 = 6000 m².

Question 17: Correct Option (a)

Area = &frac1;2 × 24 × 10 = 120 cm². Side = √(12^2 + 5^2) = √(144 + 25) = √169 = 13 cm.

Question 18: Correct Option (c)

LCM(6, 9, 15) = 90 = 2 × 3^2 × 5. To form a square, multiply by 2 × 5 = 10. Smallest square = 90 × 10 = 900.

Question 19: Correct Option (b)

(a^2 - b^2) / (a + b) = (a - b)(a + b) / (a + b) = a - b.

Question 20: Correct Option (b)

SP = Marked Price × (1 - Discount/100) = 1500 × 0.80 = ■1200.

Question 21: Correct Option (b)

Sum of digits = 2 + 1 + y + 5 = 8 + y. For 8 + y to be divisible by 9, y = 1.

Question 22: Correct Option (b)

Move decimal 11 places to the left: 1.496 × 10^11 m.

Question 23: Correct Option (b)

TSA = 2(lb + bh + hl) = 2(80 + 48 + 60) = 2(188) = 376 cm².

Question 24: Correct Option (c)

Outcomes = {HH, HT, TH, TT}. Outcomes with at least one head = {HH, HT, TH} (3 outcomes). P = 3/4.

Question 25: Correct Option (b)

x + (x + 1) + (x + 2) = 51 ⇒ 3x + 3 = 51 ⇒ 3x = 48 ⇒ x = 16.

Question 26: Correct Option (c)

400^2 = 160,000. It ends with 4 zeros.

Question 27: Correct Option (b)

Inverse proportion: 6 × 12 = 9 × d ⇒ d = 72 / 9 = 8 days.

Question 28: Correct Option (a)

Combine like terms: 7x² + (-4x + 9x) + (5 - 10) = 7x² + 5x - 5.

Question 29: Correct Option (b)

The additive inverse of -x is +x. Hence, additive inverse of -11/19 is +11/19.

Question 30: Correct Option (b)

Sum of interior angles = (n - 2) × 180° = (6 - 2) × 180° = 4 × 180° = 720°.

Question 31: Correct Option (b)

Gain = 575 - 500 = ■75. Gain % = (75 / 500) × 100 = 15%.

Question 32: Correct Option (b)

256 = 2^8 = 2^6 × 2^2 = 64 × 4. Dividing by 4 leaves 64 = 4^3 (a perfect cube).

Question 33: Correct Option (a)

In coordinate geometry (x, y), x is the abscissa (3) and y is the ordinate (-5).

Question 34: Correct Option (b)

Radius r = 14 cm. Area = πr² = (22/7) × 14 × 14 = 22 × 28 = 616 cm².

Question 35: Correct Option (a)

Using identity a^2 - b^2 = (a - b)(a + b): (5x)² - (7y)² = (5x - 7y)(5x + 7y).

Question 36: Correct Option (c)

(2/3)^(-2) = (3/2)^2 = 9/4.

Question 37: Correct Option (b)

For 1331: Sum of odd place digits = 1 + 3 = 4; sum of even place digits = 3 + 1 = 4. Difference = 0.

Question 38: Correct Option (b)

SP = 800 × (1 - 10/100) = 800 × 0.9 = ■720.

Question 39: Correct Option (c)

x/2 - x/3 = 1/4 + 1/5 ⇒ x/6 = 9/20 ⇒ x = 54/20 = 2.7.

Question 40: Correct Option (c)

Area = &frac1;2 × d1 × d2 ⇒ 120 = &frac1;2 × 10 × d2 ⇒ 5 d2 = 120 ⇒ d2 = 24 cm.

Question 41: Correct Option (b)

180 = &frac1;2 × (18 + 12) × h ⇒ 15 h = 180 ⇒ h = 12 cm.

Question 42: Correct Option (c)

Amount = 2000(1 + 10/100)^2 = 2000(1.1)² = 2000 × 1.21 = ■2420.

Question 43: Correct Option (c)

(5^2)^3 ÷ 5^4 = 5^6 ÷ 5^4 = 5^(6-4) = 5^2 = 25.

Question 44: Correct Option (b)

Square numbers end only in 0, 1, 4, 5, 6, or 9. They never end in 2, 3, 7, or 8.

Question 45: Correct Option (c)

The highest power of the variable x in the expression is 3.

Question 46: Correct Option (c)

Range = Maximum value - Minimum value = 30 - 8 = 22.

Question 47: Correct Option (c)

Volume = πr²h ⇒ 1540 = (22/7) × r² × 10 ⇒ r² = 49 ⇒ r = 7 cm.

Question 48: Correct Option (b)

Higher speed results in less time taken, so speed and time are inversely proportional.

Question 49: Correct Option (a)

x² + (4 + 10)x + (4 × 10) = x² + 14x + 40.

Question 50: Correct Option (a)

Multiplication is always distributive over addition for rational numbers: a(b + c) = ab + ac.

SECTION B: INTELLIGENCE (Q. 51 - 75) [25 Qs x 2 Marks = 50 Marks]

Question 51: Correct Option (b)

A doctor treats a patient; a teacher instructs a student.

Question 52: Correct Option (b)

Differences: +4, +6, +8, +10, +12. 30 + 12 = 42.

Question 53: Correct Option (a)

Shift each letter by +1: D→E, E→F, L→M, H→I, I→J = EFMIJ.

Question 54: Correct Option (d)

Copper, Zinc, and Iron are metallic elements; Wood is a non-metallic organic material.

Question 55: Correct Option (b)

C is mother of A. D is father of C ⇒ D is grandfather of A, so A is granddaughter of D.

Question 56: Correct Option (a)

East 10 km and West 10 km cancel out. Remaining displacement = 5 km South.

Question 57: Correct Option (b)

Letters decrease by -2 in position: Z(26), X(24), V(22), T(20), R(18), P(16).

Question 58: Correct Option (b)

Position from end = (30 - 12) + 1 = 19th.

Question 59: Correct Option (a)

Shift each letter by +6: M+6=S, O+6=U, Q+6=W = SUW.

Question 60: Correct Option (b)

18 × 12 ÷ 4 + 5 - 6 = 18 × 3 + 5 - 6 = 54 + 5 - 6 = 53.

Question 61: Correct Option (b)

Favourable outcomes: {5, 6} (2 outcomes). P = 2/6 = 1/3.

Question 62: Correct Option (d)

Digits reverse order (1 8 5) and laterally invert.

Question 63: Correct Option (b)

Pattern: a^2 + b^2 = c^2. 8^2 + 15^2 = 64 + 225 = 289.

Question 64: Correct Option (b)

In a 6-person circle, the person opposite position 2 (Q) is position 5 (T).

Question 65: Correct Option (a)

Pencils and Erasers are disjoint subsets within the set of Stationary.

Question 66: Correct Option (c)

January, March, and May have 31 days. April has 30 days.

Question 67: Correct Option (c)

Pattern: n^2. 8^2 = 64. Similarly, 10^2 = 100.

Question 68: Correct Option (b)

Sum of letter positions: P(16) + I(9) + G(7) = 32.

Question 69: Correct Option (b)

Hour hand at 9 and minute hand at 12 form a right angle = 90°.

Question 70: Correct Option (b)

4 small 1x1 squares + 1 large 2x2 square = 5 squares.

Question 71: Correct Option (b)

Letters available: P, R, E, P, A, R, A, T, I, O, N can spell P-A-P-E-R.

Question 72: Correct Option (c)

Series of squares of consecutive natural numbers: 1^2, 2^2, 3^2, 4^2, 5^2, 6^2, 7^2 = 49.

Question 73: Correct Option (a)

Hierarchical order: Member (1) → Family (2) → Community (3) → Society (5) → Nation (4).

Question 74: Correct Option (a)

Intersection of books and pens lies inside pencils ⇒ Some books are pencils (I follows). II is invalid.

Question 75: Correct Option (a)

Letter 'E' is horizontally symmetric, so its top and bottom reflection is identical to 'E'.

SECTION C: ENGLISH (Q. 76 - 100) [25 Qs x 2 Marks = 50 Marks]

Question 76: Correct Option (b)

According to the passage, Dr. B.R. Ambedkar is widely regarded as the chief architect of the Indian Constitution.

Question 77: Correct Option (c)

Dr. Ambedkar served as the Chairman of the Drafting Committee of the Constituent Assembly.

Question 78: Correct Option (b)

The passage opens by stating that the Constitution of India is the supreme law of the land.

Question 79: Correct Option (b)

The passage states that the Constitution assures its citizens justice, equality, and liberty.

Question 80: Correct Option (a)

'Fundamental' means serving as a necessary base or core principle.

Question 81: Correct Option (a)

Past Perfect Continuous tense ('had been studying') is used for an action completed before another past event.

Question 82: Correct Option (a)

Passive form of 'Who + V2 + Obj?' is 'By whom + was/were + Obj + V3?'

Question 83: Correct Option (b)

Universal truths or scientific facts retain present tense in Indirect Speech.

Question 84: Correct Option (a)

'Honest' has a silent 'h' producing vowel sound /o/ ('an'); 'small' has a consonant sound ('a').

Question 85: Correct Option (b)

'Among' is used when referring to distribution among three or more entities.

Question 86: Correct Option (a)

'Yet' is a coordinating conjunction showing contrast between two clauses.

Question 87: Correct Option (a)

'May' is used for asking formal permission polite speech.

Question 88: Correct Option (a)

When two singular nouns express a single combined idea or dish, they take a singular verb ('is').

Question 89: Correct Option (a)

A negative main statement ('haven't finished') takes a positive question tag ('have you?').

Question 90: Correct Option (b)

'Courteous' means polite, respectful, or considerate in manner.

Question 91: Correct Option (b)

'Ancient' means belonging to the distant past; 'modern' means relating to the present.

Question 92: Correct Option (a)

A 'panacea' is a universal cure or solution for all diseases or problems.

Question 93: Correct Option (b)

'Once in a blue moon' means something that happens very rarely or infrequently.

Question 94: Correct Option (b)

The correct spelling is 'Receive' (following 'e' after 'c').

Question 95: Correct Option (b)

'News' is an uncountable singular noun requiring singular verb 'is' instead of 'are'.

Question 96: Correct Option (c)

Arrangement: 'Prevention is better than cure (R, S), health is wealth (P, Q).'

Question 97: Correct Option (a)

A group of ships or naval vessels operating together is called a 'fleet'.

Question 98: Correct Option (a)

Positive degree structure: 'No other [noun] is as [adj] as [subject]'.

Question 99: Correct Option (a)

The male counterpart of a spinster (unmarried woman) is a 'bachelor'.

Question 100: Correct Option (b)

'By leaps and bounds' means progressing or growing very rapidly and dramatically.

SECTION D: GENERAL SCIENCE (Q. 101 - 125) [25 Qs x 2 Marks = 50 Marks]

Question 101: Correct Option (b)

Friction is a contact force that acts in direction opposite to motion.

Question 102: Correct Option (b)

A barometer is an instrument used to measure atmospheric pressure.

Question 103: Correct Option (b)

Lubricants form a smooth layer reducing direct surface contact and friction.

Question 104: Correct Option (d)

Sound requires a material medium for propagation and cannot travel through vacuum.

Question 105: Correct Option (b)

Loudness ∝ (Amplitude)^2; doubling amplitude increases loudness by 4 times.

Question 106: Correct Option (b)

Lemon juice contains citric acid which dissociates into ions, conducting electricity.

Question 107: Correct Option (c)

Dispersion occurs when white light passes through a prism splitting into VIBGYOR.

Question 108: Correct Option (b)

Hypermetropia (farsightedness) is corrected using a convex lens.

Question 109: Correct Option (b)

Jupiter is the largest planet, with mass greater than all other planets combined.

Question 110: Correct Option (b)

A gold-leaf electroscope detects the presence and nature of electric charge.

Question 111: Correct Option (b)

Teflon has high heat resistance and low friction, preventing oil/water from sticking.

Question 112: Correct Option (b)

Zinc is a reactive metal above hydrogen in reactivity series: $Zn + 2HCl \rightarrow ZnCl2 + H2\uparrow$.

Question 113: Correct Option (b)

Graphite has layered structure with free delocalised electrons that conduct electricity.

Question 114: Correct Option (b)

Compressed Natural Gas (CNG) burns completely producing very low emissions.

Question 115: Correct Option (b)

Dark zone contains unburnt carbon vapours and is the least hot part of the flame.

Question 116: Correct Option (b)

CFCs release chlorine radicals in stratosphere which decompose ozone (O3) molecules.

Question 117: Correct Option (b)

Threshing loosens grain from stalks; winnowing then separates grain from chaff.

Question 118: Correct Option (b)

Sir Alexander Fleming discovered Penicillin in 1928 from Penicillium notatum fungus.

Question 119: Correct Option (b)

Endemic species are restricted to a particular region (e.g. Lion-tailed macaque in Western Ghats).

Question 120: Correct Option (b)

Chlorophyll in chloroplasts absorbs light energy for photosynthesis.

Question 121: Correct Option (b)

Frogs and fish release eggs and sperms into water where external fertilization occurs.

Question 122: Correct Option (b)

Pituitary gland attached to brain secretes hormones regulating other glands.

Question 123: Correct Option (b)

Mitochondria generate ATP energy molecules through cellular respiration.

Question 124: Correct Option (b)

Hydrated ferric oxide (rust) forms in presence of both atmospheric oxygen and moisture.

Question 125: Correct Option (a)

Edward Jenner developed the smallpox vaccine in 1796.

SECTION E: SOCIAL SCIENCE (Q. 126 - 150) [25 Qs x 2 Marks = 50 Marks]

Question 126: Correct Option (b)

Battle of Buxar took place on 22 October 1764 between EIC and combined forces of Mir Qasim, Shuja-ud-Daula, and Shah Alam II.

Question 127: Correct Option (a)

Bahadur Shah Zafar II was declared symbolic leader and Emperor of Hindustan in May 1857.

Question 128: Correct Option (a)

Henry Derozio, a professor at Hindu College Calcutta, inspired youth to question social traditions.

Question 129: Correct Option (b)

General Dyer opened fire on an unarmed peaceful gathering at Jallianwala Bagh in Amritsar.

Question 130: Correct Option (b)

Bal Gangadhar Tilak gave this historic slogan during the Indian freedom struggle.

Question 131: Correct Option (b)

Renewable resources (solar, wind, water) replenish naturally within reasonable timeframes.

Question 132: Correct Option (b)

Ploughing along contour lines forms natural barriers against water flow down slopes.

Question 133: Correct Option (c)

Anthracite is hard coal with over 80-90% carbon content and highest heating value.

Question 134: Correct Option (c)

Sericulture refers to silk farming and rearing of Bombyx mori silkworms.

Question 135: Correct Option (b)

Ahmedabad in Gujarat is termed Manchester of India due to its vast textile mill industry.

Question 136: Correct Option (a)

Preamble outlines India as Sovereign, Socialist, Secular, Democratic Republic.

Question 137: Correct Option (c)

Article 32 allows citizens to move Supreme Court to enforce fundamental rights.

Question 138: Correct Option (c)

Rajya Sabha is a permanent House; members serve a term of 6 years with 1/3rd retiring every 2 years.

Question 139: Correct Option (c)

The Supreme Court located in New Delhi is the apex court of appeal in India.

Question 140: Correct Option (a)

Minimum Wages Act protects workers from exploitation by fixing statutory minimum wages.

Question 141: Correct Option (b)

Thomas Munro extended Ryotwari settlement in Madras and Bombay presidencies.

Question 142: Correct Option (b)

Birsa Munda led the 'Ulgulan' movement in Chota Nagpur plateau (Jharkhand) in late 1890s.

Question 143: Correct Option (b)

Natural gas is trapped above crude oil layers in petroleum reservoirs.

Question 144: Correct Option (a)

Natural growth rate = Birth rate - Death rate.

Question 145: Correct Option (b)

Government is constitutionally obligated to ensure basic public facilities accessible to all.

Question 146: Correct Option (a)

Swami Dayanand Saraswati established Arya Samaj advocating 'Back to the Vedas'.

Question 147: Correct Option (b)

Bauxite is a clay-like mineral from which alumina and aluminum are extracted.

Question 148: Correct Option (b)

Under Article 53, executive power of the Union is vested in the President of India.

Question 149: Correct Option (a)

AICC passed Quit India resolution at Gowalia Tank Maidan in Bombay on 8 August 1942.

Question 150: Correct Option (b)

The 86th Constitutional Amendment Act 2002 inserted Article 21A making education a fundamental right.